

Lecture-8

Mathematics 1 (15B11MA111)

Module: Jacobians

CO1 [C105.1]

Topics to be covered

➤ **Jacobians and its properties**

➤ **Examples**

➤ **Reference for the lecture**

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Jacobian

➤ Definition

If u and v function of two independent variables x and y , then

the determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ is called the Jacobian of u, v with

respect to x and y . It is defined as $J(u, v)$ or $\frac{\partial(u,v)}{\partial(x,y)}$.

For three variables: It is defined as

$$J(u, v, w) \text{ or } \frac{\partial(u,v,w)}{\partial(x,y,z)} \text{ or } \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Property-1: If u and v are function of r and s , and r, s are function of x and y then

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,s)} \cdot \frac{\partial(r,s)}{\partial(x,y)}$$

Proof: Since u and v are composite functions of x and y

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial x} = u_r r_x + u_s s_x$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \frac{\partial s}{\partial y} = u_r r_y + u_s s_y$$

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial v}{\partial s} \frac{\partial s}{\partial x} = v_r r_x + v_s s_x$$

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial s} \frac{\partial s}{\partial y} = v_r r_y + v_s s_y$$

$$\begin{aligned}
\frac{\partial(u,v)}{\partial(r,s)} \cdot \frac{\partial(r,s)}{\partial(x,y)} &= \begin{vmatrix} u_r & u_s \\ v_r & v_s \end{vmatrix} \cdot \begin{vmatrix} r_x & r_y \\ s_x & s_y \end{vmatrix} \\
&= \begin{vmatrix} u_r r_x + u_s s_x & u_r r_y + u_s s_y \\ v_r r_x + v_s s_x & v_r r_y + v_s s_y \end{vmatrix} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} \\
&= \frac{\partial(u,v)}{\partial(x,y)}
\end{aligned}$$

Property-2 (Chain rule): If J_1 is the jacobian of u, v with respect to x and y and J_2 is the jacobian of x, y with respect to u and v then $J_1 \cdot J_2 = 1$ i.e. $\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1$.

Proof: Let $u = u(x, y)$ (1)

and $v = v(x, y)$ (2)

So that u and v are functions of x and y .

Differentiate equations (1) and (2) partially with respect to u and v , respectively we get

$$1 = \frac{\partial u}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial u} = u_x x_u + u_y y_u \quad (\text{with respect to } u)$$

$$0 = \frac{\partial u}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial v} = u_x x_v + u_y y_v \quad (\text{with respect to } v)$$

$$0 = \frac{\partial v}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial u} = v_x x_u + v_y y_u \quad (\text{with respect to } u)$$

$$1 = \frac{\partial v}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \frac{\partial y}{\partial v} = v_x x_v + v_y y_v \quad (\text{with respect to } v)$$

Now

$$\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} \cdot \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$$

$$= \begin{vmatrix} u_x x_u + u_y y_u & u_x x_v + u_y y_v \\ v_x x_u + v_y y_u & v_x x_v + v_y y_v \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Property-3: If functions u, v, w of three independent variables x, y, z are not independent then the Jacobian of u, v, w with respect to x, y, z vanishes, i.e. $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 0$.

Property-4: Jacobian of implicit functions

If u_1, u_2, u_3 are the implicit functions of x_1, x_2, x_3 i.e.

$$F_1(u_1, u_2, u_3, x_1, x_2, x_3) = 0$$

$$F_2(u_1, u_2, u_3, x_1, x_2, x_3) = 0$$

$$F_3(u_1, u_2, u_3, x_1, x_2, x_3) = 0$$

then

$$\frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} = (-1)^3 \frac{\frac{\partial(F_1, F_2, F_3)}{\partial(x_1, x_2, x_3)}}{\frac{\partial(F_1, F_2, F_3)}{\partial(u_1, u_2, u_3)}}.$$

Property-5 (Functional relationship)

Let $u_1, u_2, u_3, \dots, u_n$ are the implicit functions of $x_1, x_2, x_3, \dots, x_n$ then the necessary condition for the existence of a relation of the form $F_1(u_1, u_2, u_3, \dots, u_n) = 0$ is that the Jacobian $\frac{\partial(u_1, u_2, u_3, \dots, u_n)}{\partial(x_1, x_2, x_3, \dots, x_n)}$ should vanish i.e.

$$\frac{\partial(u_1, u_2, u_3, \dots, u_n)}{\partial(x_1, x_2, x_3, \dots, x_n)} = 0$$

Example-1: If $x = r\cos\theta, y = r\sin\theta$, find $\frac{\partial(x,y)}{\partial(r,\theta)}$ and $\frac{\partial(r,\theta)}{\partial(x,y)}$.

Solution: We know $\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$

Using $x = r\cos\theta, y = r\sin\theta$, we get,

$$\frac{\partial x}{\partial r} = \cos\theta, \frac{\partial x}{\partial \theta} = -r\sin\theta \text{ and } \frac{\partial y}{\partial r} = \sin\theta, \frac{\partial y}{\partial \theta} = r\cos\theta$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

$$\text{and } \frac{\partial(r,\theta)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix}$$

Using $x = r\cos\theta, y = r\sin\theta$, we get,

$$x^2 + y^2 = r^2, \tan\theta = \frac{y}{x}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \frac{\partial r}{\partial y} = \frac{y}{r} \text{ and } \frac{\partial \theta}{\partial x} = -\frac{y}{r^2}, \frac{\partial \theta}{\partial y} = \frac{x}{r^2}$$

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} \\ -\frac{y}{r^2} & \frac{x}{r^2} \end{vmatrix} = \frac{1}{r}$$

Example-2: If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$. Show that $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2 \sin \theta$, and $\frac{\partial(r,\theta,\phi)}{\partial(x,y,z)}$.

Example-3: Calculate the Jacobian $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ of the following

(i) $u = x + 2y + z, v = x + 2y + 3z, w = 2x + 3y + 5z$.

(Answer: 2)

(ii) $u = xyz, v = xy + yz + zx, w = x + y + z$

(Ans: $(x-y)(y-z)(z-x)$)

Example-4: Verify the chain rule if $x = u, y = u \tan v, z = w$.

$$\text{Solution: } J_1 = \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} 1 & 0 & 0 \\ \tan v & u \sec^2 v & 0 \\ 0 & 0 & 1 \end{vmatrix} = u \sec^2 v$$

$$\text{Now, } u = x, \tan v = \frac{y}{u} = \frac{y}{x}, v = \tan^{-1} \frac{y}{x}, w = z.$$

$$J_2 = \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} 1 & 0 & 0 \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} & 0 \\ 0 & 0 & 1 \end{vmatrix} = \frac{x}{x^2+y^2} = \frac{1}{u \sec^2 v}$$

$$\Rightarrow J_1 J_2 = 1$$

THANK YOU

